

MICROCONTROLLER COURSE PROJECT

Design and Development of an Affordable Refreshable Braille Display Using an Embedded Control System

Abstract

We propose the design and development of an affordable refreshable Braille display using an embedded microcontroller-based control system and an alternative tactile actuation approach. The system is intended to convert digital text into dynamically refreshable Braille characters, providing visually impaired users with a reusable reading interface. The proposed design investigates a cost-effective actuation mechanism, embedded control electronics, and modular Braille cell architecture to reduce manufacturing complexity while maintaining usability. The prototype is expected to employ an embedded controller, tactile actuation mechanism, user interface circuitry and power management components. The project aims to improve accessibility, promote inclusive education and demonstrate an affordable embedded assistive technology platform. Future enhancements may include wireless connectivity, multilingual support and integration with educational software.

MICROCONTROLLER COURSE PROJECT

Design and Development of an Embedded Intelligent Bearing Fault Early-Warning System

Abstract

We propose the design and development of an embedded predictive maintenance system capable of continuously monitoring vibration characteristics of rotating machinery. The system will utilize vibration sensing, embedded signal processing and intelligent fault analysis to identify early indications of bearing degradation before catastrophic failure occurs. The proposed prototype will consist of an embedded controller, vibration sensor module, local alert mechanism and monitoring interface. By enabling condition-based maintenance, the system aims to reduce equipment downtime and maintenance costs while demonstrating an affordable industrial monitoring solution. Future enhancements may include wireless monitoring, edge AI diagnostics and cloud-based maintenance analytics.

MICROCONTROLLER COURSE PROJECT

Design and Development of a Self-Correcting Smart Soil Monitoring System

Abstract

We propose the design and development of an intelligent embedded soil monitoring system capable of identifying gradual sensor drift and maintaining reliable soil moisture estimation over long-term operation. The system combines environmental sensing with adaptive embedded algorithms to improve irrigation decisions without frequent manual recalibration. The prototype is expected to integrate soil moisture sensing, environmental sensing, embedded processing and user notification modules. The proposed solution aims to improve water conservation, increase measurement reliability and support precision agriculture using affordable embedded technology. Future enhancements may include weather integration, automated irrigation control and AI-assisted crop analytics.

MICROCONTROLLER COURSE PROJECT

Design and Development of an Embedded Firmware Integrity and Secure Boot Verification System

Abstract

We propose the design and development of an embedded cybersecurity platform that verifies firmware authenticity before device startup. The system will investigate secure boot principles, firmware integrity verification and trusted execution using embedded hardware. The prototype is expected to integrate an embedded controller, secure firmware verification logic, storage components and status indication modules. The proposed solution aims to strengthen device trust, prevent unauthorized firmware execution and demonstrate practical embedded cybersecurity techniques suitable for IoT and industrial systems. Future enhancements may include secure firmware updates, hardware roots of trust and advanced cryptographic techniques.

MICROCONTROLLER COURSE PROJECT

Design and Development of an Intelligent Cold Chain Integrity Monitoring System

Abstract

We propose the design and development of an embedded monitoring system for temperature-sensitive products during transportation and storage. The system continuously evaluates multiple environmental parameters to determine transport integrity and provide timely alerts when unsafe conditions are detected. The prototype is expected to integrate environmental sensing, embedded processing, local data logging and notification capabilities. The project aims to improve quality assurance in healthcare and food logistics through an affordable embedded monitoring platform. Future enhancements may include wireless fleet monitoring, predictive spoilage estimation and cloud-based logistics management.